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ORIGINAL ARTICLE

Sympathetic Ophthalmia after Proton Beam Irradiation for Choroidal Melanoma

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ABSTRACT

Purpose: To determine the incidence and to discuss pathogenic mechanisms of sympathetic ophthalmia in patients treated for choroidal melanoma.

Design: Retrospective

Methods: We analyzed the clinical and pathological data of 4867 patients, treated for choroidal melanoma at Institute Curie-Orsay, between 1998 and 2011. Were involved patients with uveitis of the adelphe eye filling clinical and angiographic criteria of sympathetic ophthalmia. Extensive work-up was undertaken to rule out another etiology of the inflammation.

Results: Three patients developed sympathetic ophthalmia respectively 7, 4 and 3 years after protontherapy. The incidence of this complication was 6.1 per 10000 patients treated by proton beam radiation. Ocular perforating injury was ruled out and radioactive effect of treatment was incriminated in the occurrence of sympathetic ophthalmia.

Conclusion: Sympathetic ophthalmia remains a rare complication which must be treated promptly and aggressively to prevent blindness. Proton beam irradiation could induce disruption of uveal tissue which enhances auto immune reactions.

KEYWORDS: Choroiditis, proton beam irradiation, sympathetic ophthalmia, uveal melanoma, uveitis

Sympathetic ophthalmia (SO) is an autoimmune inflammatory disease, which usually comes after an intraocular surgery or a penetrating ocular traumatism. Its pathogenesis consists in an autosensitization against an antigenic protein from the uvea or retina. The association of sympathetic ophthalmia with proton beam irradiation for choroidal melanoma is rarely reported in the literature.

In this study, we aimed to assess the incidence and to describe pathological mechanisms and clinical and therapeutic features of this complication in a referral center of tumor ocular diseases.

METHODS

We analyzed retrospectively clinical data of 4867 patients treated for choroidal melanoma with proton beam irradiation at the Institute Curie-Orsay

between 1998 and 2011. We included patients who developed uveitis of the other eye and who met clinical and angiographic criteria of sympathetic ophthalmia. Laboratory tests, including syphilitic serology, angioconvertase enzyme, tuberculin intradermoreaction, and interferon-gamma release assay (IGRA) and chest X rays, were performed to disclose differential diagnosis. We excluded patients with symptoms compatible with Vogt Koyonagi Harada disease or with a history of ocular injury or surgery before a diagnosis of choroidal melanoma was made.

RESULTS

Three patients who received proton beam irradiation developed sympathetic ophthalmia of the untreated eye respectively 7, 4, and 3 years after choroidal melanoma treatment. The incidence

of this complication was 6.1 per 10,000 at the Curie-Orsay proton therapy center. All patients' surgical technique and clip positioning records excluded a perioperative scleral perforation during tantulum rings suture. Our patients were referred to the Pitie Salpetriere Hospital ophthalmology department for uveitis exploration and treatment. Follow-up average time after SO therapy enhancement ranged from 6 months to 3 years.

Case 1

Mr. C. G., 56 years old, was referred by his ophthalmologist in 1998 for a large choroidal juxta-papillary melanoma complicated with a localized retinal detachment of the left eye. He underwent a proton beam irradiation then transpupillary thermotherapy (TTT) to reduce the risk of long-term radiation side effects. The treatment was successful with progressive decrease in tumor size assessed by both ultrasonography and orbital magnetic resonance imaging (MRI). MRI showed a superior choroidal scar of 2.9 mm thickness. Three years later, the patient developed neovascular glaucoma, and left eye phthisis occurred after 10 months under topical anti-glaucomatous and nonsteroid anti-inflammatory drops.

On March 2006, the patient presented with a history of right eye reduced vision and metamorphosis. Examination found vision reduced to 20/100, anterior granulomatous uveitis, a vitreous flare, and a juxta-macular neovascular membrane complicating a multifocal choroiditis. Both fluorescein angiography and OCT confirmed the presence of an extrafoveal choroidal neovascularization (Figures 1, 2). Treatment consisted of 3 intravenous flashes of corticosteroids (500 mg/d for 3 days) relayed by 1 mg/kg oral steroids associated to 150 mg/day of azathioprin, local steroids, focal laser treatment, and multiple intravitreal anti-VEGF injections. Best-corrected visual acuity after cataract surgery done in 2009 was 20/40.

Case 2

In 2003, Mrs. M. R., 63 years old, complained of right scotoma for 3 weeks. Examination found an infero-temporal pigmented mass and retinal detachment. A choroidal melanoma was highly suspected. The tumor was treated by proton beam irradiation and TTT of the scar. The patient maintained a good vision, and tumor size was stable until 18 months after therapy when she developed painful neovascular glaucoma. Enucleation of the right blind eye was performed on May 2006. Histological study confirmed the diagnosis of a malignant melanoma without scleral exteriorization.

One year later, the patient developed SO of the left eye with vision reduced to 20/200, granulomatous hypertensive uveitis, multifocal choroiditis, and vasculitis. Extensive workup failed to disclose another etiology. She was treated with 3 intravenous flashes of corticosteroids relayed by 1 mg/kg oral steroids associated with topic steroids and antiglaucomatous eyedrops. Evolution was marked by a good recovery of visual acuity (20/20), pigmented granulomatous precipitates, and pigmentary epithelium migrations.

Case 3

Mrs. D. J., 75 years old, was referred on October 2007 by her ophthalmologist for right eye pigmented supramacular mass of 7.1 mm thickness. She was treated by proton beam radiotherapy. The treatment was successful concerning tumor size reduction but was complicated by irradiation retinopathy and neovascular glaucoma requiring multiple therapeutic procedures, including panretinal photocoagulation, intravitreal anti-VEGF injection, and cyclocryotherapy. Ten months after cyclocryotherapy, her right eye became phthisical but no recurrence was detected on imaging.

In August 2011, the patient complained of blurred vision and the examination showed left panuveitis with yellowish subretinal spots at the posterior pole along with subretinal detachments. FFA showed early

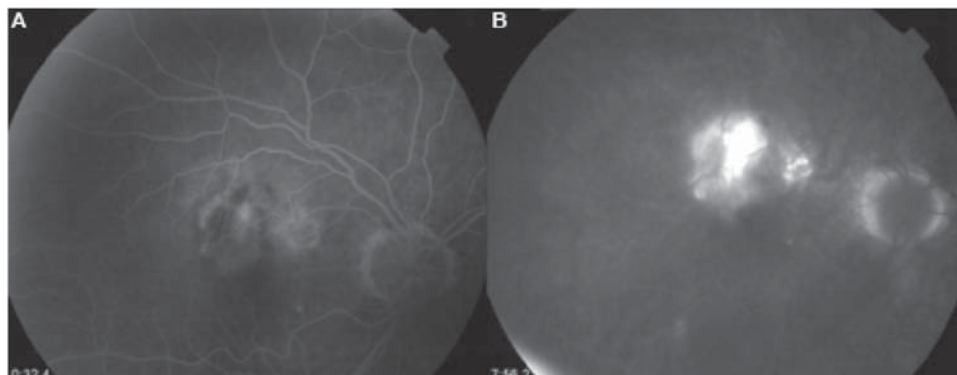


FIGURE 1 CASE 1: Fluorescein angiography shows early hyperfluorescence of the choroidal neovascular membrane (a) and late leakage (b) with inferomacular of choroidal spots.

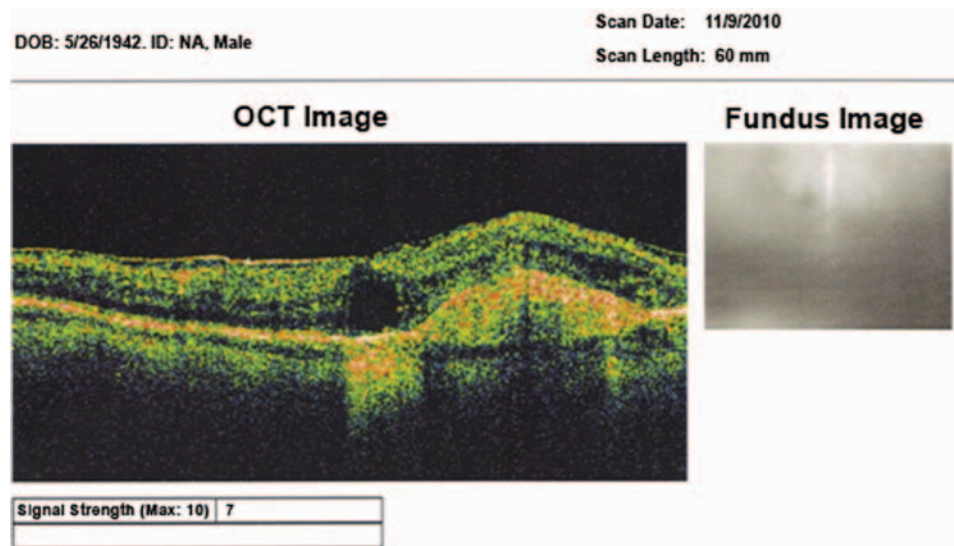


FIGURE 2 Optical coherence tomography scan of the same eye shows an area of subretinal hyperreflectivity corresponding to the choroidal neovascular membrane associated with retinal thickening and intraretinal cystic changes, and epiretinal membrane.

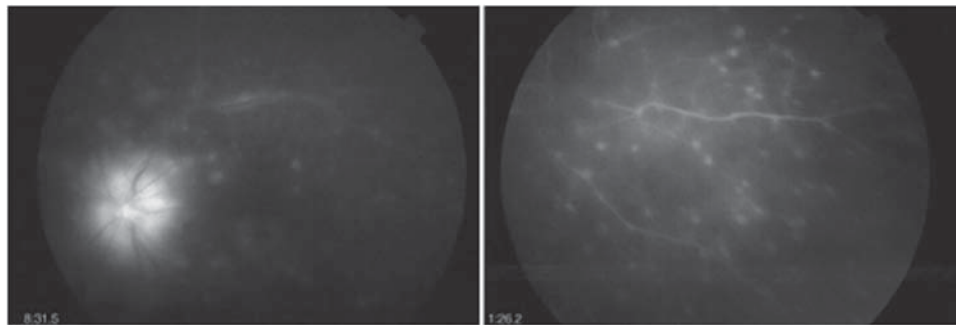


FIGURE 3 CASE 3: Fluorescein angiography of the left eye shows staining of choroidal spots, retinal vascular leakage, and optic disc hyperfluorescence.

hypofluorescence followed by late staining of the yellow spots associated to papillary hyperfluorescence and macular staining (figure 3). The patient was treated with intensive systemic steroids, local steroids, and first-line cyclophosphamide monthly. Six months after treatment, vision progressed from 20/100 to 20/40 with regression of ocular inflammation and macular edema.

DISCUSSION

Sympathetic ophthalmia is an autoimmune disease in which both peripheral blood and vitreous T lymphocytes have been shown to be reactive to retinal antigen stimulation. There may also be a genetic predisposition to the development of SO. A significant association between SO and HLA-DRB1 $\times$ 04 and -DQB1 $\times$ 04 has been reported.¹ There are a number of isolated reports of sympathetic ophthalmia associated with malignant melanoma. In most cases, a penetrating wound or the tumor extraocular extension produced disruption of the uveal tract.²⁻⁴ Orbital magnetic resonance excluded extraocular extension for our patients, and sutures that

secure tantulum rings to the sclera did not penetrate the uvea. In the second patient, histology of the enucleated globe excluded a microscopic extrascleral extension.

The findings of our cases further support the hypothesis that there is a risk of developing sympathetic ophthalmia after charged-particle therapy in the absence of a penetrating injury of the uveal tract. Disruption of uveal tissue can occur after proton beam⁵ and helium ion irradiation⁶ and ruthenium plaque brachytherapy resulting from the radioactive effect of the treatment.⁷ Expression of melanocytic antigens secondary to this uveal disruption may have induced sympathetic ophthalmia in our patients. T-cell responses specific for melanoma and melanocytes appear to be involved in the development of autoimmune reactions in patients with SO and VKH disease.⁸ Proton beam irradiation of uveal melanoma has great advantages because of the homogenous dose delivered to the tumor and the possibility of sparing normal tissue close to the tumor. This could explain the very low incidence of SO (0.06%) in our study.⁹

In case 3, beside proton beam irradiation, other procedures used in this patient might cause

sympathetic ophthalmia. The incidence of sympathetic ophthalmia is 5.8 and 0.67% after noncontact and contact Nd:YAG cyclotherapy, respectively.¹⁰ Some authors described sympathetic ophthalmia following laser cyclocoagulation confirmed by histologic and immunohistologic study showing migration of uveal pigments to the limbic conjunctiva with lymphocytic infiltration around it in the exciting eye.^{11–15}

In our patients, aggressive therapy combining a high dose of corticosteroids and first-line immunosuppressive agent was efficient to limit inflammation and prevent blindness. Even if patients had poor initial visual acuity (20/200 to 20/100), a good recovery of vision was obtained (20/40 to 20/20). This study shows that SO should be treated promptly and aggressively¹⁶ and that anti-VEGF injections can be efficient to treat choroidal neovascularization secondary to SO.¹⁷

CONCLUSION

Specialists should be aware that SO following proton beam radiotherapy is extremely rare. However, this complication, which affects the remaining eye, could lead to blindness if appropriate therapy is not promptly started.

Declaration of interest: The authors report no conflicts of interest. The authors alone are responsible for the content and writing of the paper.

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